

Dirty Cafe Sales: Exploratory Data Analysis & Revenue Optimization

- **Project Overview**

Analyzed transactional sales data from a retail cafe operation to identify primary revenue drivers, evaluate sales channels, and improve data collection practices.

- **Core Objective**

Determine whether sales volume, product pricing, dining location, or payment method has the greatest impact on total cafe revenue.

- **Data Hygiene Summary**

1. Numerical Imputation: Imputed missing values in numerical fields(Quantity, Price Per Unit) Using column means to preserve statistical distribution
2. Categorical Imputation: Categorized 7% of missing metadata(Location, Payment Method) as "Unspecified" rather than dropping rows, preserving 100% of financial transaction value.
3. Standardization & Formatting: Rounded correlation coefficients and summary metrics to 2 decimal places to minimize visual clutter and follow reporting standards.
4. Categorical Encoding: Converted raw categorical codes into clear text labels for clean reporting and chart visualizations.

- **Bottom Line**

Revenue is overwhelmingly driven by basket size(number of items per order) and unit pricing, while customer spending remains channel-neutral across payment methods and dining types.